MIL-P-24351A(SH)
27 August 1985
SUPERSEDING
MIL-P-24351(SHIPS)
24 September 1968
(See 6.6)

MILITARY SPECIFICATION

PRIMER COATING, ALKYD, BLUE (FORMULA NO. 6N35-2)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a blue coating for use as an after-pickling, anticorrosive primer on HY-130 steel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-R-266 - Resin Alkyd; Solutions.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking of.

MILITARY

MIL-A-15206 - Aluminum Stearate, Technical.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Document: https://assist.dla.mil - Downloaded: 20261019T21:29Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods for Sampling and Testing.
FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.
FED-STD-595 - Colors.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 93 - Standard Test Method for Flash Point by Pensky-Martens Closed Tester. (DoD adopted)
D 185 - Standard Test Method for Coarse Particles in Pigments, Pastes, and Paints. (DoD adopted)
D 235 - Standard Specification for Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent). (DoD adopted)
D 476 - Standard Specification for Titanium Dioxide Pigments.
D 562 - Standard Test Method for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 563 - Standard Test Method for Phthalic Anhydride Content of Alkyd Resins and Resin Solutions. (DoD adopted)
D 605 - Standard Specification for Magnesium Silicate Pigment (Talc).
D 609 - Standard Methods for Preparation of Steel Panels for Testing Paint, Varnish, Lacquer, and Related Products.
D 870 - Standard Method of Water Immersion Test of Organic Coatings on Steel. (DoD adopted)
D 1210 - Standard Test Method for Fineness of Dispersion of Pigment Vehicle Systems. (DoD adopted)
D 1296 - Standard Test Method for Odor of Volatile Solvents and Diluents. (DoD adopted)
D 1364 - Standard Test Method for Water in Volatile Solvents (Fischer Reagent Titration Method). (DoD adopted)
D 1394 - Standard Methods for Chemical Analysis of White Titanium Pigments. (DoD adopted)
D 1475 - Standard Test Method for Density of Paint, Varnish, Lacquer, and Related Products. (DoD adopted)
D 1542 - Standard Method of Qualitative Tests for Rosin in Varnishes. (DoD adopted)
D 1640 - Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings at Room Temperature.
D 1729 - Standard Practice for Visual Evaluation of Color Differences of Opaque Materials. (DoD adopted)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

ASTM (Continued)
D 2244 - Standard Method for Instrumental Evaluation of Color Differences of Opaque Materials. (DoD adopted)
D 2369 - Standard Test Method for Volatile Content of Coatings. (DoD adopted)
D 2371 - Standard Test Method for Pigment Content of Solvent-Type Paints.
D 2372 - Standard Method of Separation of Vehicle of Solvent-Type Paints.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT
Rules and Regulations - Rule 102

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Composition. The coating formulation (for Navy Standard formula no. 6N35-2) shall consist of ingredients conforming to the applicable specifications in the proportions specified in table I. Whenever formula no. 6N35-2 is specified, the enamel shall conform to this specification.

TABLE I. Coating formulation for formula no. 6N35-2.

Ingredients	Specification	Mass 1/
Titanium dioxide 2/	ASTM D 476, type IV	55.0
Barium metaborate 3/		330.0
Phthalocyanine blue pigment 4/		5.5
Magnesium silicate 6/	ASTM D 605	83.0
Aluminum stearate	MIL-A-15206	6.5
Alkyd resin solution	TT-R-266, type I, class A	382.0
Petroleum spirits	ASTM D 235, type IV	230.0
Driers 5/		

1/ This formula is in parts by mass. Use of kilograms as mass units results in a volume slightly in excess of 835 liters. Use of pounds as mass units results in a volume slightly in excess of 100 gallons.

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:29Z
Check the source to verify that this is the current version before use.

2/ The titanium dioxide content of this pigment shall be not less than 90 percent by mass.
3/ Busan 11-M1, Buckman Laboratories, Inc., or equivalent as approved by the Naval Sea Systems Command (NAVSEA).
4/ Phthalocyanine blue pigment or a NAVSEA approved equivalent. The 5.5 mass units shall be considered only as a guide. Actual amount of pigment shall be adjusted to conform to color as specified (see 3.5.2).
5/ The type and amount of driers used shall be the choice of the manufacturer. The driers shall be lead-free and the resulting paint shall conform to dry-time as specified (see 3.5.1).
6/ Magnesium silicate used for this formulation shall be certified asbestos free. Paint manufacturer shall obtain certification to this effect from the raw material supplier.

3.2 Manufacture. The component raw materials shall be mixed and ground as required, to produce a product which is uniform, suitable for the purpose intended, and in full conformity with the requirements of this specification.

3.2.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.5) and in accordance with proper safety and application procedures. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity.

3.3 Volatile portion. The volatile portion of the enamel shall conform to the following requirements by volume (see 4.5):

(a) Aromatic compounds with eight or more carbon atoms except ethylbenzene: 8 percent maximum.
(b) Ethylbenzene and toluene: 20 percent maximum.
(c) Compounds with olefinic or cycloolefinic type of unsaturation: 1 percent maximum.
(d) Branched ketones: 20 percent maximum.
(e) Total of (a), (b), (c) and (d): 20 percent maximum.

3.4 Formulated coating requirements.

3.4.1 Quantitative requirements. The coating shall conform to the quantitative requirements specified in table II, when tested as specified (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

TABLE II. Quantitative requirements.

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent mass of coating	42.5	46.5
Volatiles, percent mass of coating	28.5	32.5
Nonvolatile vehicle, percent mass of coating (calculated by difference)	23.0	27.0
Lead (percent)	----	0.06
Water, percent mass of coating	----	0.5
Coarse particles and skins, percent mass of coating retained on 325 sieve	----	0.1
Consistency, Krebs-Stormer 200 r/min		
Grams	140	190
Krebs units (equivalent)	70	80
Density		
Kilograms per liter	1.28	1.33
Pounds per gallon	10.7	11.1
Fineness of dispersion, Hegman units	4	----
Phthalic anhydride, percent mass of nonvolatile vehicle	23.0	----
Titanium dioxide, percent mass of pigment	10.0	----
Barium metaborate ($BaB_2O_4.H_2O$), percent mass of pigment 1/	60.0	----
Flash point		
Degrees Celsius	38	----
Degrees Fahrenheit	100	----

1/ Determinations by the barium and boron methods (see 4.4.4) shall agree within 2 percent to ensure proper composition of this pigment.

3.4.2 Condition in container. The coating shall be free from grit, seediness, skinning or livering in a freshly opened full container and shall show no more pigment settling or caking than can be readily redispersed with a paddle to a uniform state, when tested as specified (see 4.4).

3.4.3 Odor. The odor of the coating shall be characteristic of the volatiles permitted, when tested as specified (see 4.4).

3.4.4 Rosin and rosin derivatives. Rosin and rosin derivatives shall not be present in the coating when tested as specified (see 4.4).

3.4.5 Phenolic resins. Phenolic resins shall not be present in the coating when tested as specified (see 4.4).

3.4.6 Reducibility and dilution stability. The packaged material as received, shall be tested in accordance with 4.4.6. The reduced product shall remain stable and uniform (except for pigment settling) for not less than 4 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

3.4.7 Storage stability. The product shall be readily broken up with a paddle to a smooth, uniform consistency, shall match the standard color, shall not exceed 6 hours dry-hard time, and shall not liver, curdle, gel, or show any other objectionable properties for at least 1 year after date of acceptance (see 4.4.6).

3.4.8 Skinning and gelling. The coating shall not show any gel formation, skinning, agglomeration, or other evidence of incompatibility when held in a half full 236 milliliter (mL) (1/2-pint) tinplate can, at 60 degrees Celsius (°C) (15.6 degrees Fahrenheit (°F)) and inspected at 24, 48, and 72 hours as specified in 4.4.7.

3.5 Coating application.

3.5.1 Drying time. The set-to-touch time of the applied coating shall be not more than 1 hour. The dry-hard time shall be not more than 6 hours.

3.5.2 Color. The color shall approximately match the dry standard color chip number 15200 of FED-STD-595 when tested as specified in 4.4.9.

3.5.3 Brushing properties. The coating shall be brushed out and laid off without excess drag on the brush when tested as specified in 4.4.10. When dry, the brushed surface shall be free from sags and runs, and shall show a minimum of brush marks.

3.5.4 Spraying properties. The coating shall be sprayed satisfactorily in all respects, and shall show no running, sagging, or streaking when tested as specified in 4.4.11. The film shall show no dusting, mottling, or color separation.

3.6 Performance of applied coating.

3.6.1 Flexibility. The coating shall show no evidence of cracking or flaking when subjected to the flexibility test specified in 4.4.12.

3.6.2 Resistance to water immersion. There shall be no evidence of wrinkling or blistering of the coating immediately after removal from water, no more than a slight effect after 2 hours, and no evidence of change of hardness, adhesion, or visual properties 24 hours after immersion, when the coating is tested as specified in 4.4.13.

3.7 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form OSHA-20 and is found in FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance. Quality assurance shall be provided in accordance with method 1031 of FED-STD-141 as hereinafter supplemented. Submission of forms in accordance with method 1031 shall cover complete tests other than the 1-year storage stability test (see 4.4.6). The storage stability requirement shall apply for 1 year after acceptance, regardless of other testing or prior acceptance of the material.

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of all paint of the same formula number from a single, uniform batch, or uniform blend of batches offered for delivery at one time.

4.3.2 Sampling of ingredient materials. When requested by the testing laboratory or the contracting activity, one pint of each of the ingredients in the formula specified in table I shall be supplied for test purposes.

4.4 Test procedures. The coating shall be tested in accordance with the applicable methods specified in table III, and as specified herein. Additional chemical and physical tests shall be conducted as necessary to determine that the ingredients and proportions specified in table I have been used. Failure of any sample to pass any test and noncompliance with the requirements of this specification shall be cause for rejection of the lot represented by the sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

TABLE III. Test procedures.

Tests	Applicable test method FED-STD-141	Applicable ASTM test method	Test paragraph	Requirements paragraph
Pigment content		D 2372	---	Table II
Volatiles		D 2369	---	Table II
Nonvolatile vehicle content	4053		---	Table II
Water		D 1364	---	Table II
Coarse particles		D 185	---	Table II
Consistency		D 562	---	Table II
Density		D 1475	---	Table II
Fineness of dispersion		D 1210	---	Table II
Phthalic anhydride		D 563	4.4.2	Table II
Titanium dioxide		D 1394	4.4.3	Table II
Barium metaborate, barium method		------	4.4.4.1	Table II
Barium metaborate, boron method		------	4.4.4.2	Table II
Flash point		D 93	---	Table II
Condition in container	3011		---	3.4.2
Odor		D 1296	---	3.4.3
Rosin and rosin derivatives		D 1542	---	3.4.4
Phenolic resins	5141		---	3.4.5
Reducibility and dilution stability	4203		4.4.5	3.4.6
Storage stability	3011		4.4.6	3.4.7
Skinning and gelling		----	4.4.7	3.4.8
Drying time		D 1640	4.4.8	3.5.1
Color		D 1729	4.4.9	3.5.2
Brushing properties	4321		4.4.10	3.5.3
Spraying properties	4331		4.4.11	3.5.4
Flexibility	6221		4.4.12	3.6.1
Resistance to water immersion		D 870	4.4.13	3.6.2

4.4.1 Separation of pigment and vehicle. The pigment shall be separated from the vehicle in accordance with ASTM D 2371 for analysis of phthalic anhydride, titanium dioxide, and barium metaborate contents, and for determination of the presence of rosin or rosin derivatives and phenolic resins.

4.4.2 Phthalic anhydride. A suitable portion of the vehicle shall be evaporated and collected in accordance with ASTM D 2371 on a steam bath until the volume has been reduced to approximately 10 mL and analyzed in accordance with ASTM D 563, using the correction for carbonates as specified in this method.

Source: https://assist.dla.mil - Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

4.4.3 Titanium dioxide. Titanium dioxide shall be determined in accordance with ASTM D 1394.

4.4.4 Barium metaborate. Percent barium metaborate by barium and boron determinations shall be as specified in 4.4.4.1 and 4.4.4.2.

4.4.4.1 By barium determination. Transfer a 1.0 gram sample of extracted pigment, weighed to the nearest 0.1 milligram (mg), to a 150-mL beaker. Add 10 mL concentrated nitric acid and 10 mL, 30 percent hydrogen peroxide solution. Heat until the blue color disappears. Filter and wash, collecting the filtrate in a 250-mL beaker. Dilute to about 200 mL with distilled water. Plate out copper at 2 amperes for 3 hours. Heat to boiling and add 5 mL of 1:1 sulfuric acid slowly and digest for 1 hour. Filter through a tared Gooch crucible containing two glass filter pads. Wash the precipitate with hot water containing a few drops of sulfuric acid. Finish washing with pure distilled water. Dry at 130°C for 2 hours. Cool and weigh as barium sulfate. Calculate barium as barium metaborate ($BaB_2O_4.H_2O$) by the following formula:

$$BaB_2O_4.H_2O = \frac{103.4A}{S}$$

Where: A = weight of barium sulfate precipitate in grams
S = sample weight in grams.

Report barium metaborate as percent mass of pigment.

4.4.4.2 By boron determination. Transfer a 0.5 gram sample of extracted pigment weighed to the nearest 0.1 mg to a 250-mL beaker. Add 10 mL concentrated hydrochloric acid. Heat to about 80°C for 1 hour and filter. Add three drops of methyl red indicator (0.2 gram of methyl red in 100 mL ethyl alcohol). Add 20 percent sodium hydroxide solution until the red disappears. Add 20 mL in excess. Add 1:1 sulfuric acid until the methyl red color is nearly reached. Add 0.1 N sulfuric acid until the solution turns red. The barium sulfate and lead sulfate will precipitate. Add 4 grams of mannitol and three drops of phenolphthalein indicator solution (1.0 gram phenolphthalein in 100 mL ethyl alcohol). Titrate to the phenolphthalein end point with standardized 0.1 N sodium hydroxide. (The solution starts out red, turns yellow, then red again at the end point.) Calculate boron as barium metaborate as follows:

$$BaB_2O_4.H_2O = \frac{A(N)\ (12.5)}{S}$$

Where: A = mL of sodium hydroxide solution used
N = normality of sodium hydroxide solution
S = sample size in grams.

Report barium metaborate as percent mass of pigment.

4.4.5 Reducibility and dilution stability. Reducibility and dilution stability shall be determined in accordance with method 4203 of FED-STD-141, using 50 mL of enamel and 50 mL of petroleum spirits conforming to ASTM D 235, type IV. Observe immediately after mixing and 4 hours after mixing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

4.4.6 Storage stability. The Government at its option may examine the enamel (stored unopened in its original container) for compliance with 3.4.7 at any time within 1 year of acceptance. The examination shall be in accordance with method 3011 of FED-STD-141.

4.4.7 Skinning and gelling. Skinning and gelling shall be determined as follows. Transfer 115 grams of the enamel to a 1/2-pint, friction top, tinplate can approximately 3 inches in height and diameter. The can should be about one half full. Close the can securely and store at 60 ± 2°C for inspection at 24, 48, and 72 hours. At each inspection, examine the enamel in an unstirred condition by carefully probing with a spatula. The liquid portion shall be smooth flowing and completely free of any detectable trace of gelation or skin. Do not omit inspections at 24 and 48 hours, as they contribute to the severity of the test by increasing the exposure of the enamel to air. Promptly close the can after each inspection and return to storage. At 72 hours, remove the can and allow to cool at room temperature for 1 hour, before making the final inspection. After examining the enamel in unstirred condition, hand stir with a spatula, and again examine for conformance to 3.4.8.

4.4.8 Drying time. Drying time shall be determined in accordance with ASTM D 1640, except that the film thickness shall be such as to give a dry film thickness of 1.0 ± 0.15 mils (25 ± 4 micrometers (μm)).

4.4.9 Color. A film of the enamel shall be prepared by drawing down with a doctor blade of 6 mils (150 μm) gap clearance on a plain opaque white glass panel. After a 48-hour drying period, a visual comparison shall be made with the color chip as specified in ASTM D 1729. If doubt exists as to the acceptability of the match, the difference shall be determined by instrument as specified in ASTM D 2244. The allowable color difference shall be two units.

4.4.10 Brushing properties. The coating shall be applied in accordance with method 4321 of FED-STD-141 to steel panels 3 by 1-1/2 feet (920 by 460 millimeters (mm)).

4.4.11 Spraying properties. Sixteen parts by volume of the coating shall be reduced with one part by volume of thinner conforming to ASTM D 235, type IV. The reduced coating shall be sprayed on a steel panel to a dry film thickness of 1.0 ± 0.2 mils (25 ± 6 μm) in accordance with method 4331 of FED-STD-141. For referee test, automatic application shall be used in accordance with method 2131 of FED-STD-141.

4.4.12 Flexibility. Flexibility shall be determined in accordance with method 6221 of FED-STD-141. The coating shall be applied to a flat tinplate panel of approximately 31 gauge by a doctor blade to obtain a dry film thickness of 1.0 ± 0.3 mils (25 ± 7 μm). The panel shall air-dry for 2 hours, then bake for 24 hours at 100 to 105°C. The panel shall be removed from the oven and allowed to stand for 30 minutes at 25 ± 5°C. The panel shall be bent over a 1/8 inch (3.2 mm) mandrel and the film at the bend shall be examined under a magnification of 7 diameters.

Source: https://assist.dla.mil Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

4.4.13 Resistance to water immersion. Resistance to water immersion shall be determined in accordance with ASTM D 870 using a vapor or solvent degreased type 2 or type 3 steel panel as specified in ASTM D 609. A coating film shall be drawn down to a dry film thickness of 1.0 ± 0.3 mils (25 ± 7 µm) and air dried for 48 hours. Approximately half of the coated portion of the panel shall be immersed in distilled water at 25 ± 2°C for 18 hours. The panel shall be examined immediately after removal, after 2 hours, and after 24 hours.

4.5 Certification data/report. The contractor shall certify that each lot of paint is in accordance with this specification (see 6.2.2). In addition the following information shall be included:

(a) Toxicological data and formulations required to evaluate the safety of the material for the proposed use (see 3.2.1).
(b) Certification that volatility requirements are met (see 3.3).
(c) Conformance of each lot of ingredient material, stating test results and source, as applicable (see 4.4).

4.6 Inspection of packaging. Sample packages and packs, inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions.)

5.1 Packaging, packing and marking. The enamel shall be packaged, packed and marked in accordance with PPP-P-1892. The level of packaging shall be level A, B or C, and the level of packing shall be level A, B or C, as specified (see 6.2.1). The enamel shall be furnished in 1-gallon cans or 5-gallon pails, as specified (see 6.2.1).

5.1.1 Special marking. In addition to other markings required (see 6.2.1), the containers shall be marked with the following statement: "The volatile content of this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District" (see 6.4).

6. NOTES

6.1 Intended use. This primer coating is intended for use as an identification primer for HY-130 steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing required (see 5.1).
(c) Size of container required (see 5.1).
(d) Special marking required (see 5.1.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.5	Certification data/report	UDI-A-23264	10.2.1

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3 Level B packaging. Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined that the enamel will be held in covered storage no more than 1 year from the date of initial packaging.

6.3 Composition by volume. For information only, and with the understanding that the mass-to-volume relationships of ingredients may vary slightly, an approximate composition by volume of the coating in this specification, as prepared according to table I, is specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:29Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Composition by volume.

Ingredient	Liters	Gallons
Barium metaborate	100.00	12.36
Titanium dioxide	13.92	1.67
Phthalocyanine blue pigment	3.40	0.41
Magnesium silicate	29.86	3.57
Aluminum stearate	6.44	0.77
Alkyd resin solution	397.91	47.70
Petroleum spirits	294.87	35.33
Drier	10.22	1.22
Total volume	856.62	103.03

6.4 Volatile content. Although the container marking (see 5.1.1) specifically refers to the South Coast Air Quality Management District (California), the specified enamel may be used anywhere else an enamel complying with 3.3 is allowed. This includes nearly all other air pollution control districts or similar areas, which control the emission of solvents into the atmosphere.

6.5 Basis of purchase. The specified enamel should be purchased by volume, the unit being a United States gallon of 231 cubic inches (3.785 liters) at 15.5°C.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 8010-N267)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-24351A(SH)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:29Z
Check the source to verify that this is the current version before use.